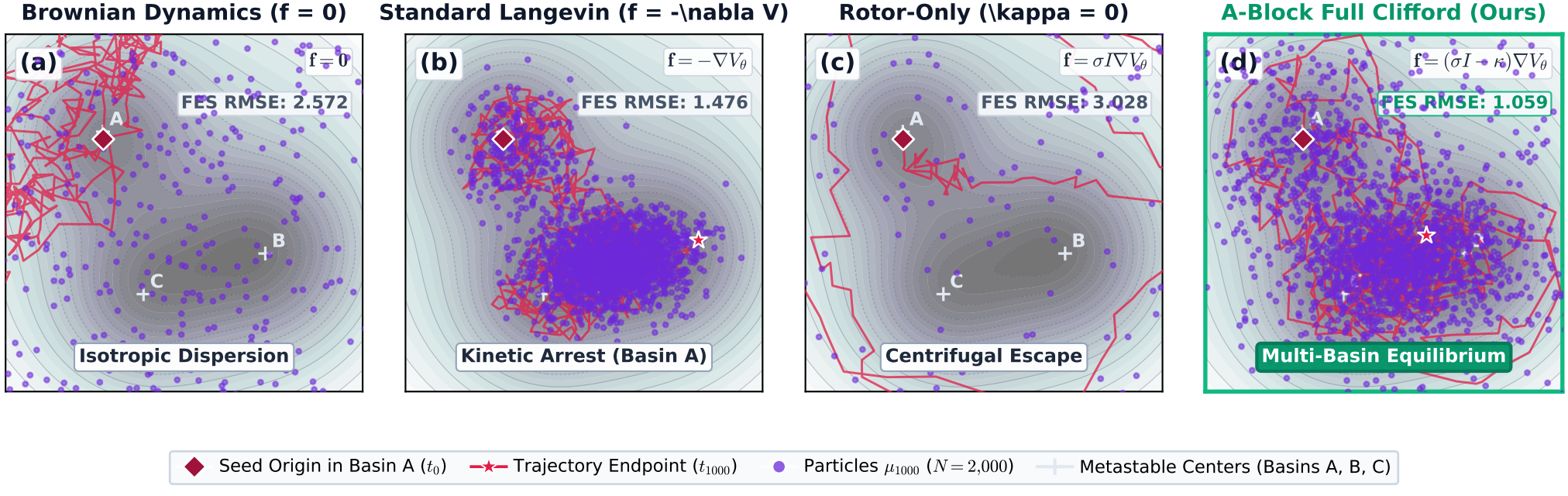

Brownian Dynamics (f = 0)
Standard Langevin (f = -\nabla V)
Rotor-Only (\kappa = 0)
A-Block Full Clifford (Ours)
(a)
$\mathbf{f} = 0$
FES RMSE: 2.572
A
B
C
Isotropic Dispersion
(b)
$\mathbf{f} = -\nabla V_\theta$
FES RMSE: 1.476
Kinetic Arrest (Basin A)
(c)
$\mathbf{f} = \sigma I \nabla V_\theta$
FES RMSE: 3.028
Centrifugal Escape
(d)
$\mathbf{f} = (\sigma I - \kappa)\nabla V_\theta$
FES RMSE: 1.059
Multi-Basin Equilibrium
Seed Origin in Basin A (t_0)
Trajectory Endpoint (t_{1000})
Particles μ_{1000} ($N = 2{,}000$)
Metastable Centers (Basins A, B, C)